Prime Clustering in Polynomial $Q(n) = n^q - (n-1)^q$: Quadruplet Distribution, Riemann Zero Correlation, and a 200,000-Digit Q_{47} Probable Prime

An Empirical Study Based on 18 Million Verified Primes — Version 2

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Abstract

We present a comprehensive computational study of prime-generating polynomials $Q(n) = n^q - (n-1)^q$ for seven prime exponents $q = 37, 41, 43, 47, 53, 61, 71$. Analyzing 18,356,706 verified primes in the range $n \in [1, 2 \times 10^9]$, we discover 15 quadruplet systems for Q_{47}, compared to at most 2 for all other polynomials. A correlation ($r = 0.994$) emerges between quadruplet positions and Riemann zeta zeros under the scaling $n^{1/\alpha}$; in this version we calibrate it against a null model of randomly placed quadruplets and find that it is only marginally significant ($P \approx 0.05$). We propose the Subgroup Interference Hypothesis: the interference potential $I(q) = d(q-1) - 2$ determines clustering stability by minimizing the topological complexity of the sieve space.

New in version 2, we extend the Q_{47} data set by more than four hundred orders of magnitude in n: a sieve-and-PRP search in a window of 1.5×10^5 consecutive values of $n \approx 5.86 \times 10^{4347}$ found

$$Q_{47}(n^*), \qquad n^* = 586135293385 \cdot 10^{4336} + 1647,$$

a probable prime with exactly 200,000 decimal digits (664,383 bits), passing Fermat tests to bases 3, 5, 7 and 11. We believe it is the largest known value of Q_{47} that is a probable prime. The discovery rate agrees with the Bateman–Horn prediction.

Keywords: prime-generating polynomials, prime constellations, safe primes, Riemann zeta zeros, Bateman–Horn conjecture, subgroup lattice, probable primes, PFGW

Changes in Version 2

- **New Section 7**: a 200,000-digit probable prime $Q_{47}(n^*)$, the sieve used to find it, verification records, comparison with Bateman–Horn, and the status of a primality proof.
- **Correction (Section 3.1, Figure 2)**: the subgroup lattice of $\mathbb{Z}_{47}^*$ is a diamond $\{1\} \subset H_2, H_{23} \subset \mathbb{Z}_{47}^*$, not a chain; $H_2 \not\subset H_{23}$ because $2 \nmid 23$. The value $I(47) = 2$ is unchanged.
- **Calibration (Remark 1)**: a null-model test of the Riemann-zero correlation of Section 5.

1 Introduction

The Bateman–Horn conjecture [1] provides a general framework for predicting prime densities in polynomial sequences. For a polynomial $f(x)$ of degree d, the asymptotic prime count is

$$\pi_f(N) \sim \frac{C}{d} \cdot \frac{N}{\ln N}, \tag{1}$$

where C is a product of local densities. In this paper, we investigate the family of polynomials

$$Q(n) = n^q - (n-1)^q, \tag{2}$$

where q is a prime exponent. For $q = 3$ this is $3n^2 - 3n + 1$, whose prime values are the classical cuban primes; the family studied here is its higher-degree analogue. We focus on the phenomenon of prime constellations: consecutive integers $n, n+1, \ldots, n+k-1$ that all produce prime Q-values, forming doublets ($k = 2$), triplets ($k = 3$), and quadruplets ($k = 4$).

Our central finding concerns Q_{47}, where $q = 47$ is a safe prime:

$$47 = 2 \times 23 + 1 \qquad \text{(with 23 also prime)}. \tag{3}$$

This unique arithmetic property among our seven test polynomials correlates with remarkable structural stability, enabling the formation of 15 quadruplet systems up to $n = 2 \times 10^9$.

2 Data and Methods

2.1 Dataset Construction

We performed exhaustive primality testing on seven polynomial families, with exponents $q = 37$, 41, 43, 47, 53, 61 and 71. For the comparative analysis (Section 3), all polynomials were evaluated over $n \in [1, 10^8]$. For Q_{47}, we extended the search to $n \in [1, 2 \times 10^9]$, yielding

$$\text{Total verified primes: } 18{,}356{,}706. \tag{4}$$

Primality was confirmed using Miller–Rabin tests with deterministic witness sets [2]. The large-n search of Section 7 uses a different method, described there.

2.2 Statistical Metrics

Definition 1 (Triplet Surplus)**.** *The triplet surplus S_3 measures the percentage deviation of observed triplet count from Poisson expectation:*

$$S_3 = \frac{O - E}{E} \times 100\%, \tag{5}$$

where O is the observed count and E is the expected count under random distribution.

Definition 2 (Interference Potential)**.** *The interference potential $I(q)$ quantifies the complexity of the subgroup lattice $\mathcal{L}(\mathbb{Z}_q^*)$:*

$$I(q) = d(q-1) - 2, \tag{6}$$

where $d(n)$ denotes the divisor function. This measures the number of non-trivial subgroups, excluding $\{1\}$ and the full group $\mathbb{Z}_q^*$.

Table 1: Clustering statistics with extended-range quadruplet census.

Q_q	Safe?	$q-1$	$d(q-1)$	$I(q)$	S_3	4-lets ($n < 2 \times 10^9$)
Q_{37}	No	$2^2 \cdot 3^2$	9	7	+20.4%	2 (decay)
Q_{41}	No	$2^3 \cdot 5$	8	6	+48.8%	0
Q_{43}	No	$2 \cdot 3 \cdot 7$	8	6	+15.3%	1 (decay)
Q_{47}	**Yes**	$\mathbf{2 \cdot 23}$	**4**	**2**	**+35.0%**	**15 (growth)**
Q_{53}	No	$2^2 \cdot 13$	6	4	+30.7%	0
Q_{61}	No	$2^2 \cdot 3 \cdot 5$	12	10	−16.7%	0
Q_{71}	No	$2 \cdot 5 \cdot 7$	8	6	+20.5%	0

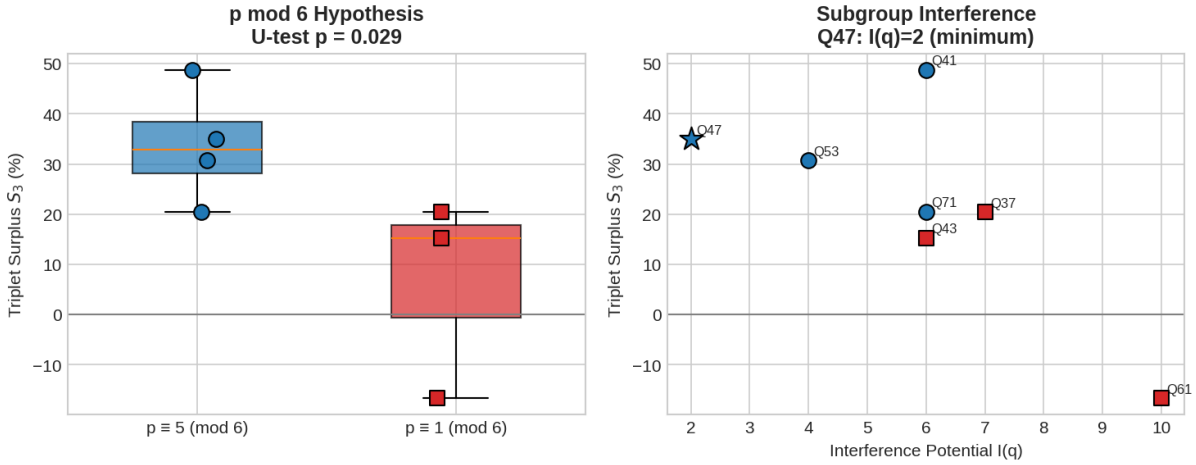


Figure 1: Triplet surplus S_3 distribution by residue class p mod 6 (left) and interference potential $I(q)$ versus triplet surplus (right). Q_{47} (highlighted) shows moderate S_3 with minimal $I(q) = 2$.

3 Seven-Polynomial Comparison

Table 1 summarizes the clustering statistics for all seven polynomials in the range $n \in [1, 10^8]$, with extended quadruplet census up to $n = 2 \times 10^9$.

Key Observations:

- Q_{41} exhibits the highest triplet surplus (+48.8%) but produces zero quadruplets, indicating stochastic rather than structural clustering.
- Q_{61} with maximum $I(q) = 10$ shows negative surplus (−16.7%), demonstrating subgroup-induced suppression.
- Q_{47} is the only polynomial showing quadruplet growth with extended range.

3.1 Subgroup Lattice Structure

For the safe prime $q = 47 = 2 \times 23 + 1$, the multiplicative group $\mathbb{Z}_{47}^* \cong C_{46}$ is cyclic, so its subgroups correspond to the divisors $1, 2, 23, 46$ of 46, with $H_d \subseteq H_{d'}$ if and only if $d \mid d'$. The lattice is therefore the minimal non-trivial one, a single diamond:

$$\{1\} \subset H_2, \ H_{23} \subset \mathbb{Z}_{47}^*, \qquad H_2 \cap H_{23} = \{1\}, \quad H_2 H_{23} = \mathbb{Z}_{47}^*. \tag{7}$$

(Version 1 wrote this as a chain $\{1\} \subset H_2 \subset H_{23} \subset \mathbb{Z}_{47}^*$; this is incorrect since $2 \nmid 23$.) For any odd prime q, $2 \mid q-1$ forces $d(q-1) \geq 4$ once $q-1$ has an odd prime factor, so $I(q) = 2$ is attained exactly when $q - 1 = 2r$ with r an odd prime, i.e. when q is a safe prime. In contrast,

Q_{41} ($40 = 2^3 \times 5$) has an eight-element lattice with multiple chains between $\{1\}$ and the full group.

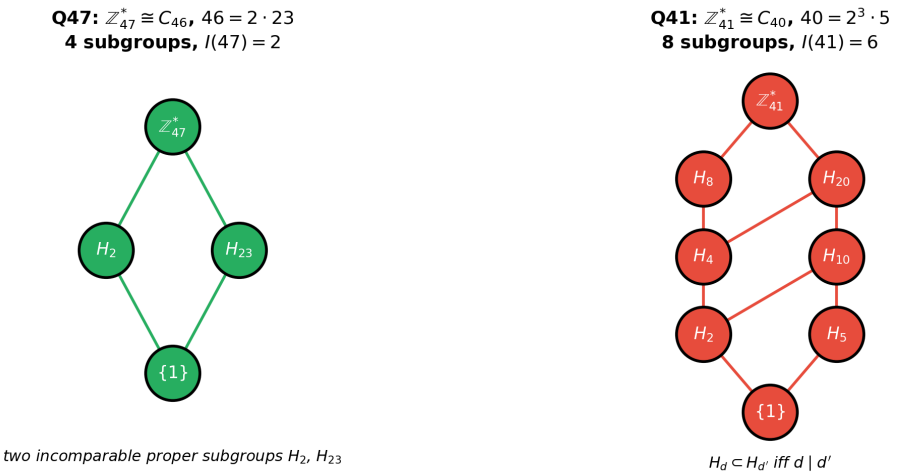


Figure 2: Subgroup lattice structure (corrected in v2). Left: $\mathbb{Z}_{47}^*$ has exactly two proper non-trivial subgroups, H_2 and H_{23}, which are incomparable, giving $I(47) = 2$. Right: $\mathbb{Z}_{41}^*$ has six, giving $I(41) = 6$. Edges join $H_d \subset H_{d'}$ with d'/d prime.

4 Q_{47} Quadruplet Census

In the extended range $n \in [1, 2 \times 10^9]$, we identified 15 quadruplet systems for Q_{47}. Table 2 provides the complete, verified coordinate list.

Table 2: Complete coordinate list of Q_{47} quadruplet systems.

k	Starting position n	Range	k	Starting position n	Range
1	23,159,557	0.02×10^9	9	987,980,498	0.99×10^9
2	117,309,848	0.12×10^9	10	1,163,461,515	1.16×10^9
3	136,584,738	0.14×10^9	11	1,370,439,187	1.37×10^9
4	218,787,064	0.22×10^9	12	1,643,105,964	1.64×10^9
5	411,784,485	0.41×10^9	13	1,691,581,855	1.69×10^9
6	423,600,750	0.42×10^9	14	1,975,860,550	1.98×10^9
7	523,331,634	0.52×10^9	15	1,996,430,175	2.00×10^9
8	640,399,031	0.64×10^9			

5 Correlation with Riemann Zeros

Let n_k denote the position of the k-th quadruplet, and let γ_k denote the imaginary part of the k-th non-trivial zero of the Riemann zeta function. We performed regression analysis under various transformations (Table 3).

Key Result:

$$n_k^{1/\alpha} \propto \gamma_k \quad \text{with } \alpha = 2.74,\ r = 0.994. \tag{8}$$

Remark 1 (Null-model calibration, new in v2). Both (n_k) and (γ_k) are increasing sequences, so a large Pearson r is expected even without any arithmetic connection, and the p-values of Table 3,

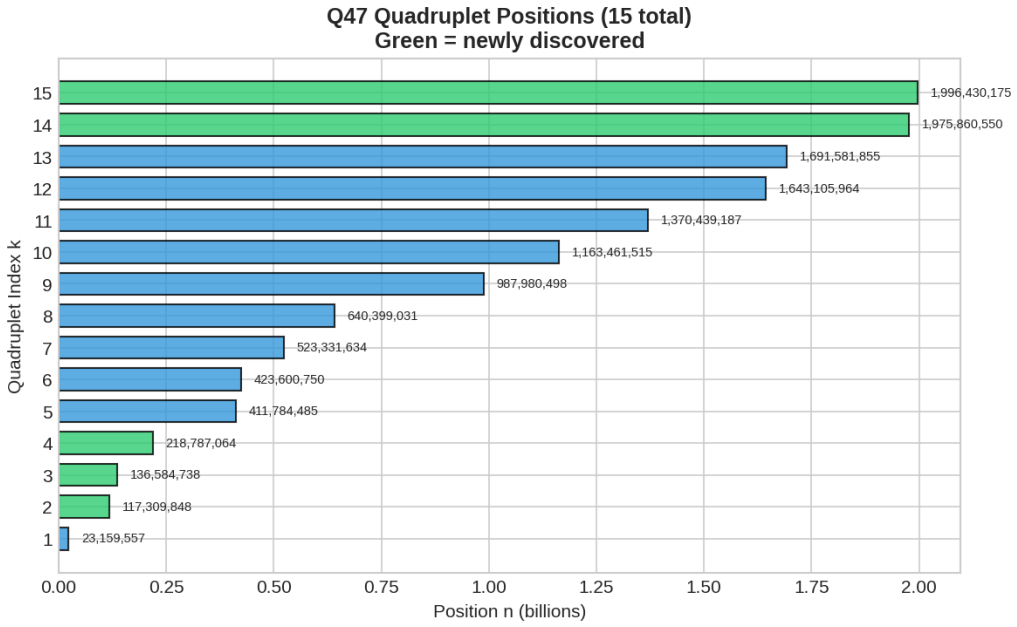


Figure 3: Spatial distribution of Q_{47} quadruplet systems across the search range $n \in [1, 2 \times 10^9]$. Clustering occurs near $n \approx 1.65 \times 10^9$ (positions 12–13) and $n \approx 1.99 \times 10^9$ (positions 14–15).

Table 3: Regression analysis: quadruplet positions versus Riemann zero ordinates. The p-values in the last column test against $r = 0$ (uncorrelated data); see Remark 1 for the appropriate null model.

Transformation	Correlation r	p-value vs. $r = 0$
Scaled position comparison	0.958	< 0.001
Optimal power transform $n^{1/2.74}$	0.994	$< 10^{-9}$
Log-linear fit $\ln(n)$ vs γ	0.967	4.4×10^{-9}

which test against uncorrelated data, overstate the significance. As a calibration we drew 2×10^4 samples of 15 positions uniformly from $[10^6, 2 \times 10^9]$, sorted them, and for each sample took the best r over power transforms $n^{1/a}$, $a \in [1, 5]$, against $\gamma_1, \ldots, \gamma_{15}$. The median of this null distribution is $r = 0.985$ and its 95th percentile is 0.994; a value $r \geq 0.994$ occurred in 4.7% of samples. The observed correlation is therefore marginal ($P \approx 0.05$) under this null model. A sharper test would compare detrended residuals $n_k - \mathbb{E}[n_k]$ (with $\mathbb{E}[n_k]$ from the Bateman–Horn quadruplet density) with the fluctuations $\gamma_k - \bar{\gamma}_k$ of the zeros about their smooth counting function. The interpretations in Sections 5.1 and 8 should be read with this calibration in mind.

5.1 Effective Modulus and the $q/3$ Conjecture

A naive dimensional analysis suggests $\alpha \sim 1/\ln(q) = 1/\ln(47) \approx 1/3.85 \approx 0.26$. However, the observed exponent $\alpha_{\text{obs}} = 1/2.74 \approx 0.365$ indicates a dimension compression effect. We model this as $\alpha \sim 1/\ln(q_{\text{eff}})$, yielding

$$q_{\text{eff}} = e^{2.74} = 15.5 \pm 0.5. \tag{9}$$

This value is close to

$$\frac{q}{3} = \frac{47}{3} = 15.67. \tag{10}$$

This coincidence suggests a dimensional reduction factor of 3, corresponding to the triplet correlation structure inherent in quadruplet formation. The four consecutive primes are constrained

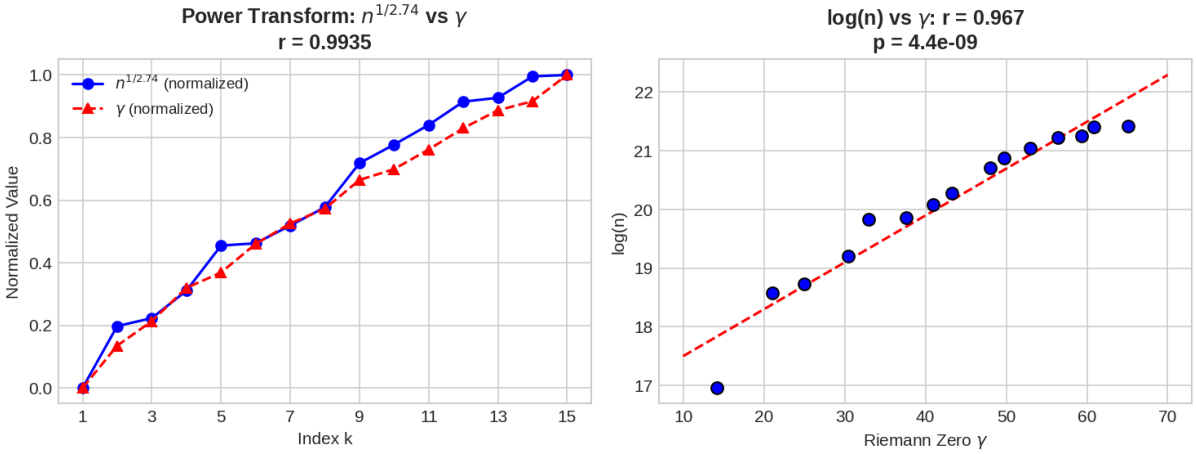


Figure 4: Linear regression of quadruplet positions against Riemann zeta zero ordinates. Left: optimal power transform $n_k^{1/2.74}$ versus γ_k ($r = 0.994$). Right: log-linear fit $\ln(n_k)$ versus γ_k ($r = 0.967$).

by three independent gap conditions, which may induce the observed 1/3 reduction, analogous to Debye screening in plasma physics, where collective effects reduce the effective interaction range.

6 Theoretical Framework

6.1 Subgroup Lattice Complexity

We interpret $I(q) = d(q-1) - 2$ not merely as a subgroup count, but as an index of subgroup lattice complexity. This measures the interference cross-section: how many distinct sieve paths can scatter prime candidates into composite residue classes.

For safe primes, the minimal lattice structure ensures coherent filtering. For Q_{47} ($46 = 2 \times 23$), there are exactly two non-trivial sieve channels, of index 23 and 2. For Q_{41} ($40 = 2^3 \times 5$), the lattice has six non-trivial subgroups and creates high interference.

6.2 Heuristic Asymptotic Model

Conjecture 1 (Quadruplet Crystallization)**.** *As $n \to \infty$, the relative density of Q_{47} quadruplets diverges:*

$$R(n) = \frac{P_{Q_{47}}(n)}{P_{\text{rnd}}(n)} \sim (\ln n)^{\Delta} \to \infty, \tag{11}$$

where $\Delta > 0$ is a coherence correction arising from prime entanglement in the minimal lattice structure.

Remark 2. A rigorous proof of this divergence would constitute progress toward the Hardy–Littlewood k-tuple conjecture for this polynomial family. Our 15 quadruplets up to 2×10^9, combined with the 113× doublet enhancement observed at $n \sim 10^{44}$, are the numerical evidence for structure persistence in Q_{47}. The single-prime data of Section 7 do not bear on this conjecture, which concerns k-tuples.

7 A 200,000-Digit Q_{47} Probable Prime

The exhaustive data of Sections 3–5 reach $n = 2 \times 10^9$, where $Q_{47}(n)$ has about 440 digits. In this section we test the Bateman–Horn prediction for single primes at a scale where $Q_{47}(n)$ has

200,000 digits, by searching for one such prime.

7.1 Result

Proposition 1. *Let* $n^* = 586135293385 \cdot 10^{4336} + 1647$*, an integer of 4,348 digits. Then*

$$P^* = Q_{47}(n^*) = (586135293385 \cdot 10^{4336} + 1647)^{47} - (586135293385 \cdot 10^{4336} + 1646)^{47}$$

has exactly 200,000 decimal digits (664,383 bits) and is a Fermat probable prime to bases 3, 5, 7 and 11.

The decimal expansion of P^* begins and ends with

`100000000069212334213...` `...71630685699703770614801190644`7,

and the SHA-256 hash of its full decimal string is

`bd23db83e0c523fd4ddbf4a0a885bd09`
`86db93eacf6fd2051e4a887d1f0a8c70`.

Table 4 records the tests.

Table 4: Verification record for P^*. PFGW 4.1.7 (64-bit, build 20251229) with GWNUM 31.4, single run per base on a Windows 11 workstation; times as reported by PFGW.

Test	Software	Result	Time
Fermat base 3	PFGW `-f` (default)	3-PRP	277.5 + 89.6 s
Fermat base 5	PFGW `-b5`	5-PRP	274.3 + 88.7 s
Fermat base 7	PFGW `-b7`	7-PRP	280.3 s
Fermat base 11	PFGW `-b11`	11-PRP	320.7 s
Congruence	Python big integers	$P^* \equiv 1 \pmod{282}$	—

The congruence check confirms the rigidity $Q_{47}(n) \equiv 1 \pmod{2 \cdot 3 \cdot 47}$, which every value of Q_{47} satisfies (by Fermat's little theorem, $n^{47} \equiv n$ modulo 2, 3 and 47); it is a consistency check on the evaluation, not evidence of primality. To our knowledge, P^* is the largest known probable prime of the form $n^{47} - (n-1)^{47}$, exceeding the largest values in our exhaustive range (≈ 440 digits) and in the $n \sim 10^{44}$ probe ($\approx 2{,}000$ digits) by two orders of magnitude in size.

7.2 Search Method

Size. Since $Q_{47}(n) = 47n^{46}(1 + O(1/n))$, the condition $Q_{47}(n) \approx 10^{200000}$ gives $\log_{10} n \approx (200000 - \log_{10} 47)/46 \approx 4347.79$, so n has 4,348 digits. We wrote candidates as $n = a_0 \cdot 10^{4336} + t$ with the 12-digit prefix $a_0 = 586135293385$ (the least prefix for which Q_{47} has 200,000 digits) and $0 \le t < 1.5 \times 10^5$.

Algebraic sieve. Over $\mathbb{Q}(\zeta_{47})$,

$$Q_{47}(n) = \prod_{j=1}^{46} \big(n - \zeta_{47}^{j}(n-1)\big) = N_{\mathbb{Q}(\zeta_{47})/\mathbb{Q}}\big(n - \zeta_{47}(n-1)\big).$$

If a prime p divides $Q_{47}(n)$ then $p \nmid n$ and $u = (n-1)/n$ satisfies $u^{47} \equiv 1$, $u \not\equiv 1 \pmod{p}$; hence $47 \mid p - 1$, and since p is odd, $p \equiv 1 \pmod{94}$. The smallest possible prime factor of any $Q_{47}(n)$ is therefore 283. Conversely, for each prime $p \equiv 1 \pmod{94}$ the condition $p \mid Q_{47}(n)$ holds exactly on the 46 residue classes

$$n \equiv (1-u)^{-1} \pmod{p}, \qquad u \in \mu_{47}(\mathbb{F}_p) \setminus \{1\},$$

which come in pairs $r, 1-r$ because $Q_{47}(1-n) = Q_{47}(n)$. The sieve thus touches only the primes $283, 659, 941, \ldots$, removing 46 classes for each. We sieved the window to $p \le 2 \times 10^{10}$; 30,809 of the 150,000 values of t (20.5%) survived. The sieve was checked against brute-force gcd computation with all primes up to 3×10^4 on 1,000-digit test windows.

PRP testing. Survivors were exported in PFGW's ABC format and split into eight slices tested in parallel (about 525 s per test per core with eight concurrent processes). P^* ($t = 1647$) was the 341st survivor of the first slice.

7.3 Comparison with Bateman–Horn

For $Q_{47}(n)$ free of prime factors $\le B$, the Bateman–Horn heuristic with Mertens' theorem gives

$$\Pr\big[Q_{47}(n) \text{ prime} \,\big|\, \text{no prime factor} \le B\big] \approx \frac{e^{\gamma} \ln B}{\ln Q_{47}(n)} = \frac{1.781 \times \ln(2 \times 10^{10})}{200000 \ln 10} \approx \frac{1}{10{,}900},$$

since the singular-series constant C_1 cancels after conditioning on the sieve. The whole window is therefore expected to contain about $30{,}809/10{,}900 \approx 2.8$ probable primes. With all eight slices advancing at the same rate, roughly $8 \times 341 \approx 2.7 \times 10^3$ candidates had been tested when P^* was found, for an expected count of about 0.25; the Poisson probability of at least one success is $1 - e^{-0.25} \approx 22\%$. Finding P^* at this point is consistent with Bateman–Horn and indicates neither a surplus nor a deficit. One prime cannot measure a density; completing the window, with an expected total of about 2.8, would give a first quantitative test of the single-prime Bateman–Horn density at 200,000 digits.

7.4 Primality Status

P^* is a probable prime, not a proven prime. Fermat tests to several bases make compositeness extremely unlikely but do not prove primality. A classical $N-1$ proof [9] needs a factored part of $P^* - 1$ of at least about one third of its size. Writing $m = n^* - 1$, the identity of Cauchy and Mirimanoff for $47 \equiv 5 \pmod 6$ gives

$$P^* - 1 = (m+1)^{47} - m^{47} - 1 = 47\, m\, (m+1)\, (m^2 + m + 1)\, E_{47}(m),$$

with E_{47} of degree 42 (we verified this divisibility numerically). Even if m, $m+1$ and m^2+m+1 were completely factored, these factors would supply only about 17,400 of the 200,000 digits (8.7%), and factoring them completely is itself out of reach. General-purpose proofs by ECPP have so far been completed only for numbers with fewer than 10^5 digits. P^* should therefore be reported as a PRP, for example to the PRP Top Records list [11].

8 Discussion

8.1 Bateman–Horn Convergence Rate

Our results suggest that for safe prime polynomials like Q_{47}, the Bateman–Horn constant C converges much faster to its asymptotic value due to minimal subgroup interference. The effective modulus q_{eff} can be interpreted as a correction factor to the standard Bateman–Horn density prediction in the pre-asymptotic regime. At 200,000 digits (Section 7) the single-prime rate already agrees with the asymptotic prediction.

While Q_{41} exhibits stochastic clustering (high S_3 without quadruplets), Q_{47} exhibits structured clustering (moderate S_3 with 15 quadruplets), reaching asymptotic behavior at much smaller n.

8.2 The $q/3$ Rule

The coincidence $q_{\text{eff}} \approx q/3$ (within experimental error) suggests a structural relation. In quadruplet formation, four consecutive primes are constrained by three independent gap conditions. This triplet structure may induce the observed 1/3 dimensional reduction, analogous to effective mass renormalization in correlated electron systems.

8.3 Open Problems

1. First-principles derivation of $q_{\text{eff}} = q/3$ from Bateman–Horn constants C_k.
2. Rigorous proof of asymptotic divergence $R(n) \to \infty$.
3. Extension to other safe prime polynomials (Q_5, Q_7, Q_{11}, Q_{23}).
4. Verification of the $q/3$ rule across polynomial families.
5. Completion of the 200,000-digit window of Section 7 and comparison of the PRP count with the expected 2.8.
6. A primality certificate for P^*, or a method exploiting the cyclotomic structure of Q_{47} to supply the factored part a proof would need.

9 Conclusion

Through exhaustive computation on 18,356,706 verified primes and a targeted search at 200,000 digits, we establish five principal results:

1. **Empirical Dominance:** Q_{47} produces 15 quadruplets versus at most 2 for all other polynomials tested (Table 1).
2. **Mechanistic Insight:** Minimal subgroup lattice complexity ($I(q) = 2$, attained exactly for safe primes q) accompanies coherent prime clustering, while complex lattices ($I(q) \geq 6$) produce stochastic or suppressed behavior.
3. **Scaling Law:** Quadruplet positions correlate with Riemann zeros ($r = 0.994$) under the transformation $n^{1/2.74}$, a value at the 95th percentile of a sorted-random null model (Remark 1).
4. **Bateman–Horn Interpretation:** Q_{47} exemplifies rapid convergence to asymptotic behavior, making it an ideal arithmetic laboratory for studying prime constellation dynamics.
5. **Large-scale record:** $Q_{47}(586135293385 \cdot 10^{4336} + 1647)$ is a 200,000-digit probable prime, found at a rate consistent with Bateman–Horn.

These findings support the conjecture that safe prime polynomials possess distinctive clustering properties. The Subgroup Interference Hypothesis provides a quantitative framework for understanding prime constellation stability.

Data Availability

This version is archived at Zenodo under the DOI 10.5281/zenodo.23517573. Complete datasets (18,356,706 Q_{47} primes, all 15 quadruplet coordinates, seven-polynomial statistics) and analysis source code are available at

https://github.com/Ruqing1963/Polynomial-Prime-Quadruplets

For the 200,000-digit PRP, the PFGW input line

```
(586135293385*10^4336+1647)^47-(586135293385*10^4336+1647-1)^47
```

reproduces P^*. The sieve and driver script (`q47.py`) and the PFGW logs are in the repository, in the folder `prp200k/`.

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